GradLoss

Grad-CAM++ 기반 주의 지도와 IOU 손실을 활용한 객체 인식 성능 개선 연구

2024학년도 PARA 겨울방학 프로젝트 보고서

GradLoss: Grad-CAM 기반 주의 지도와 IOU 손실을 활용한 객체 인식 성능 개선 연구

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GradLoss: Enhancing Object Recognition through Grad-CAM based Attention Maps and IOU Loss

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요 약

본 연구에서는 GradCAM++을 활용한 새로운 손실 함수 설계를 제안한다. GradCAM++은 딥러닝 모델의 예측 과정에서 클래스별 활성화 맵을 보다 세밀하게 시각화할 수 있는 기법으로, 모델의 해석 가능성을 향상시키는 데 유용하다. 본 연구에서는 GradCAM++에서 얻은 시각화 정보를 기반으로 손실 함수를 설계하고, 이를 학습 과정에 통합하여 모델 성능과 해석 가능성을 동시에 향상시키는 방법을 제안한다. 본 연구는 특히 복잡한 딥러닝 모델에서 해석 가능성과 정확성을 동시에 개선하는 데 기여할 것으로 기대된다.

주제어 : GradCAM, 손실 함수, 모델 해석 가능성, 딥러닝, 기계 학습

Abstract This study proposes a novel loss function design utilizing GradCAM++. GradCAM++ is a technique that can visualize class-specific activation maps more precisely during the prediction process of deep learning models, making it highly useful for enhancing model interpretability. In this study, a loss function is designed based on visualization information obtained from GradCAM++, and it is integrated into the training process to improve both model performance and interpretability. Experimental results demonstrate that the proposed loss function outperforms existing loss functions and effectively leverages GradCAM++ to contribute to the training of deep

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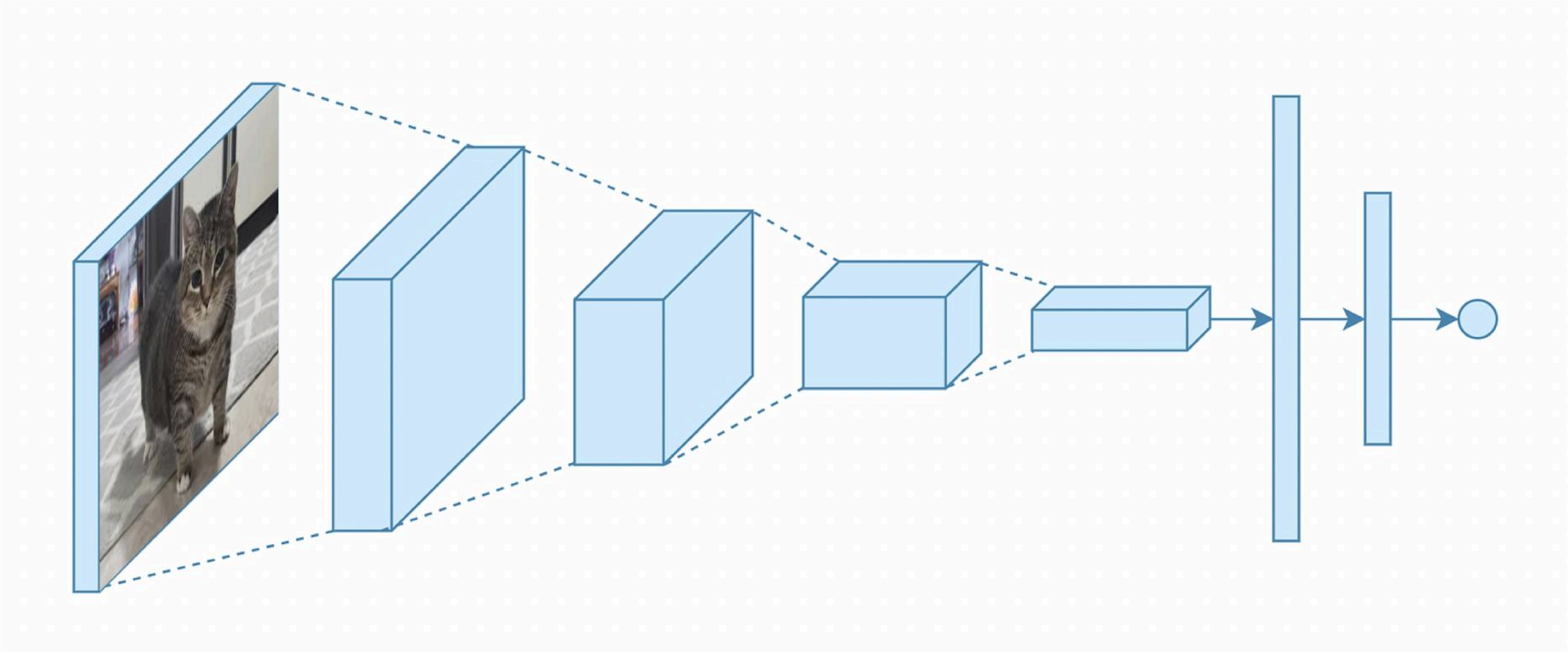
서론



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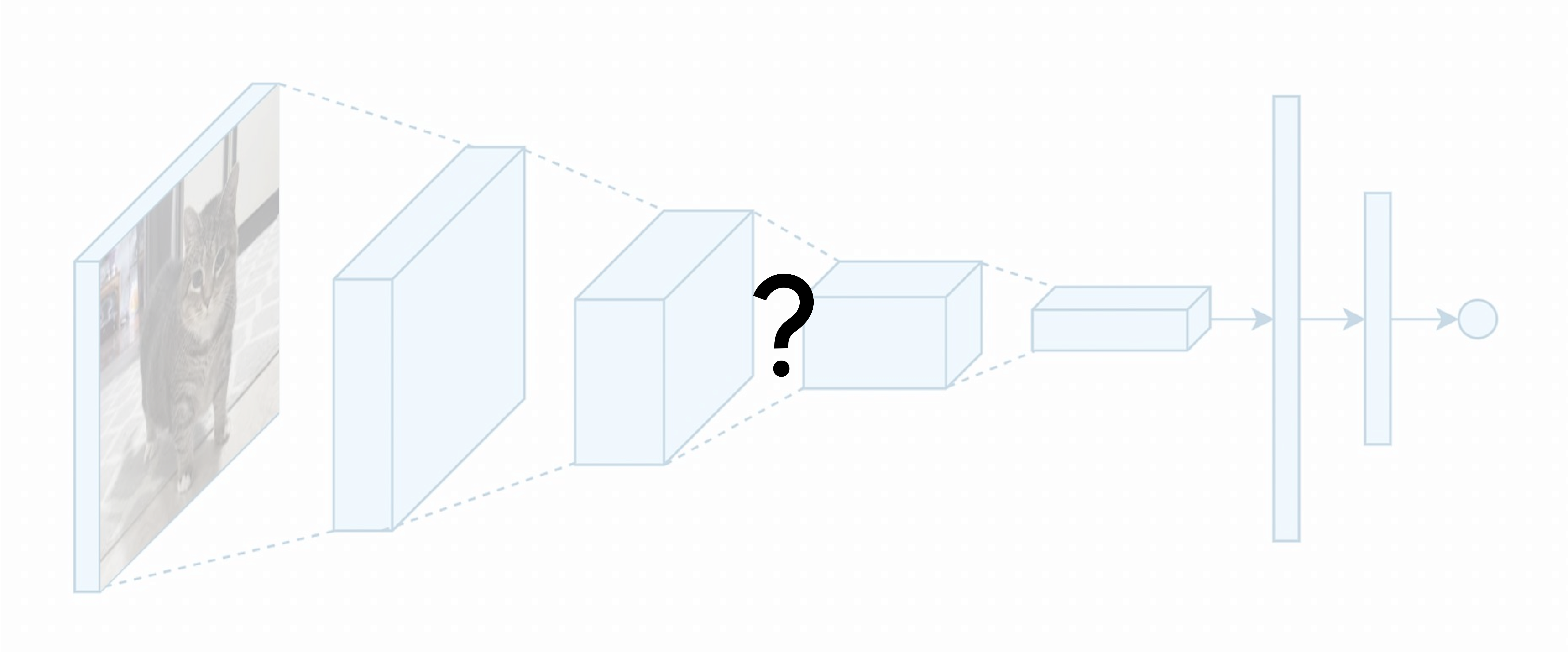

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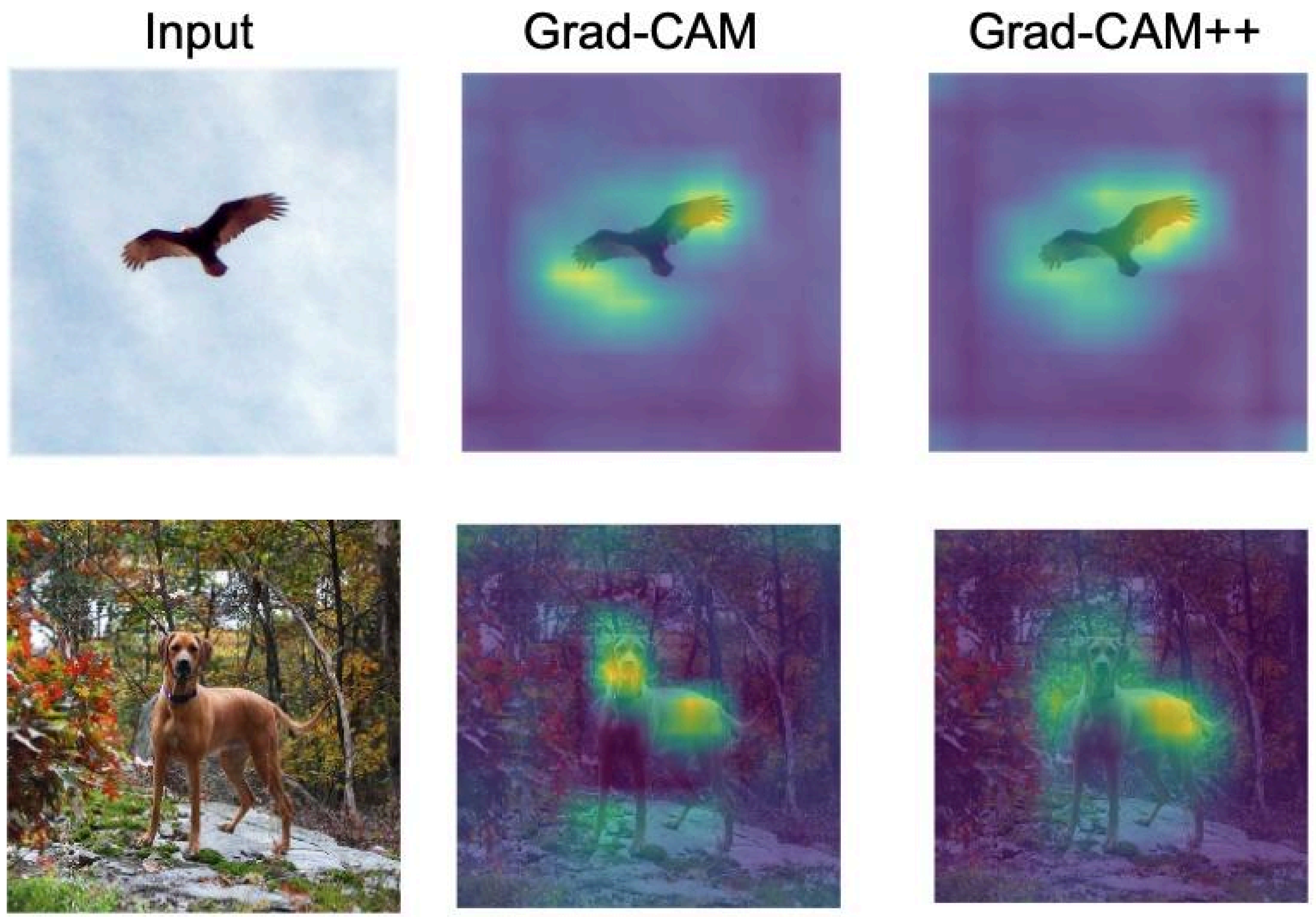
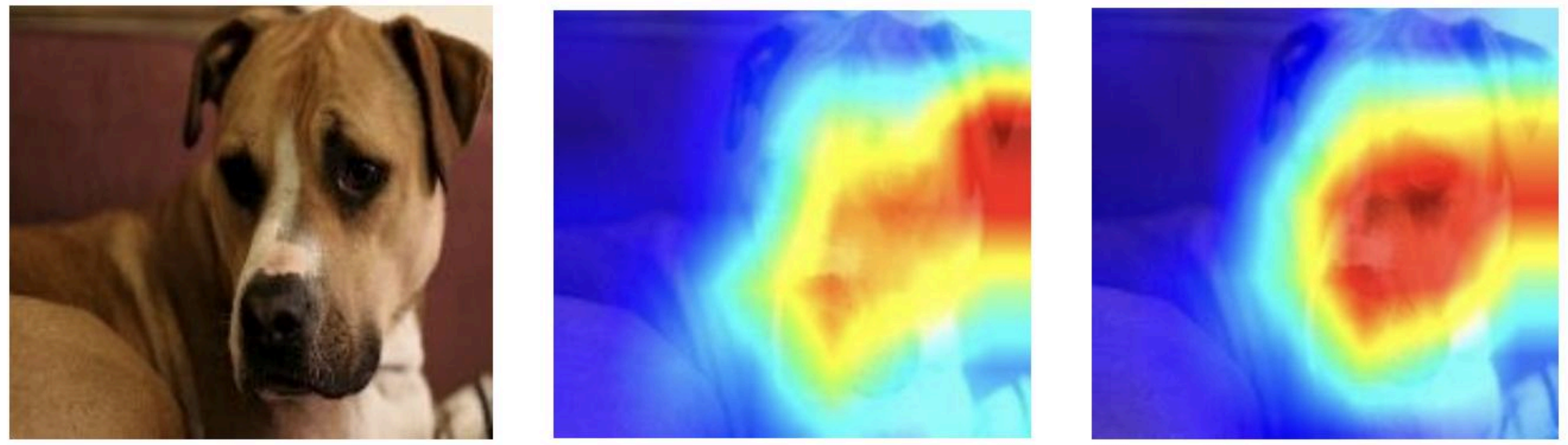
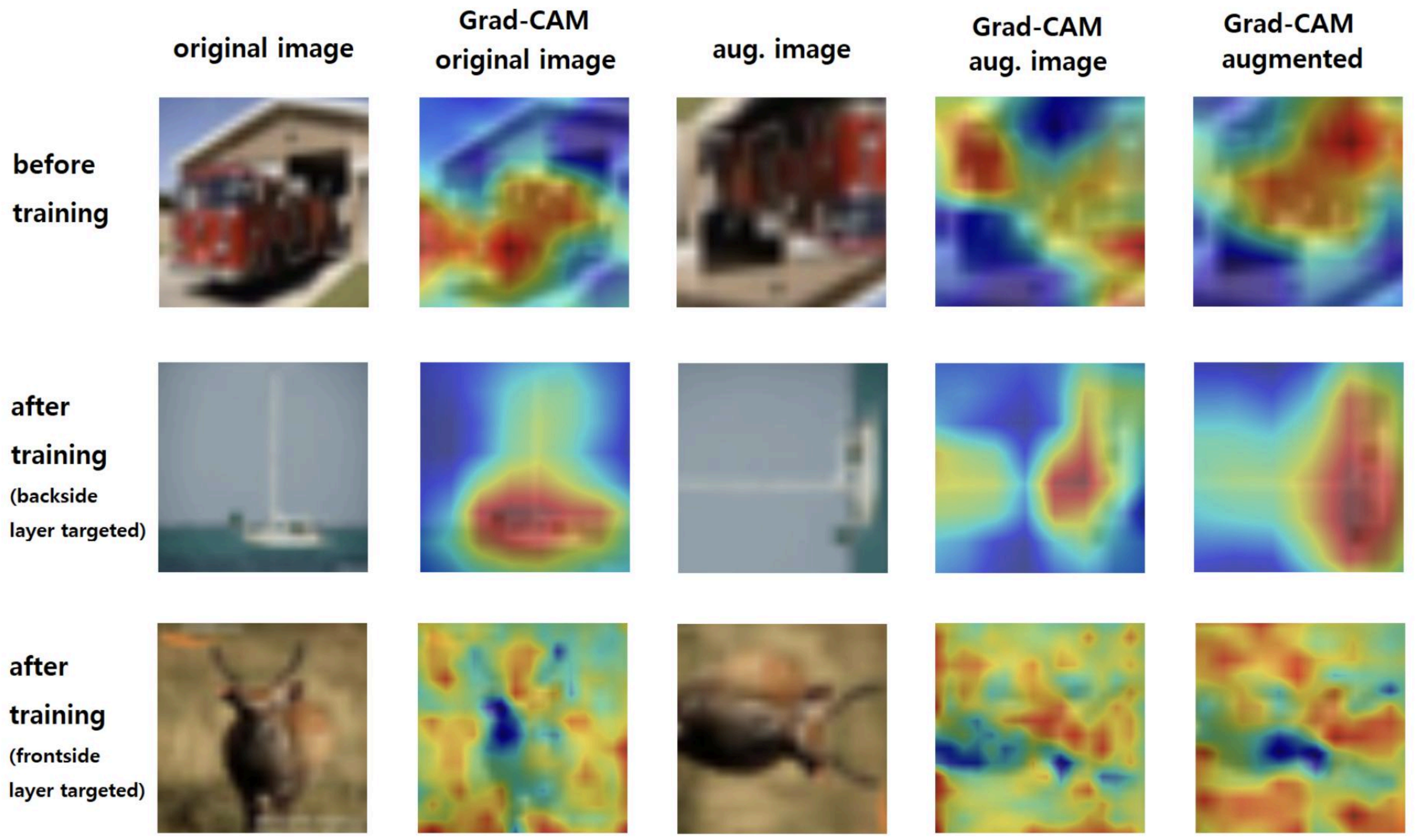


관련 연구



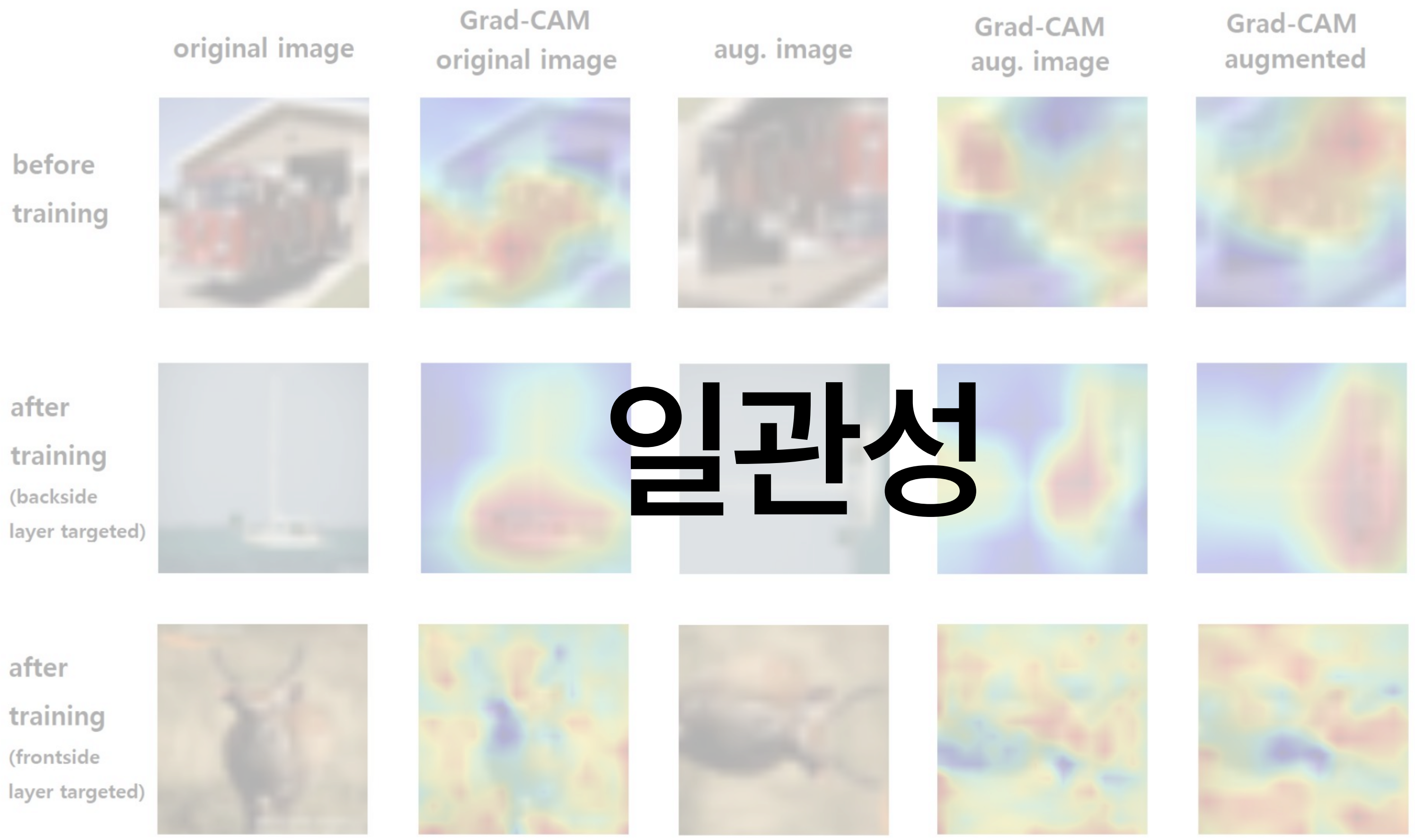
A light-weight method to foster the (Grad)CAM interpretability and explainability of classification networks



Semi-supervised Image Classification with Grad-CAM Consistency

관련 연구


특징 검출


original image
Grad-CAM
original image
aug. image
Grad-CAM
aug. image
Grad-CAM
augmented
before
training
after
training
(backside
layer targeted)
after
training
(frontside
layer targeted)
일관성

연구 설계

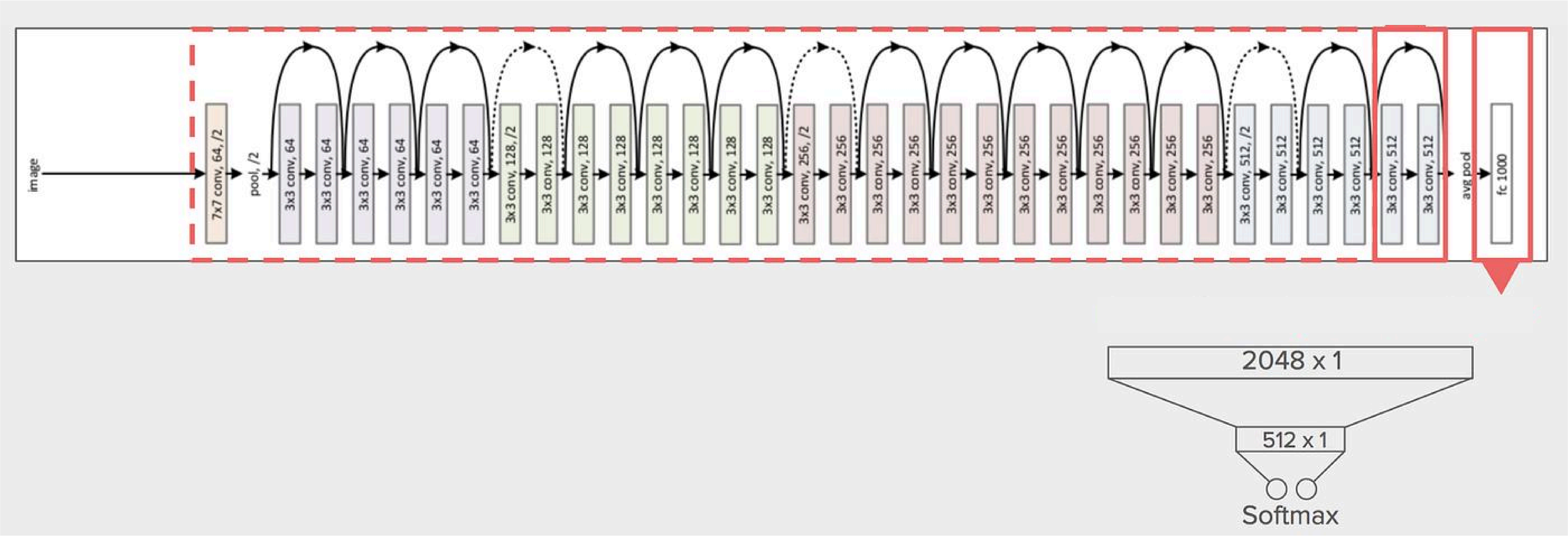


image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
avg pool
fc 1000
2048 x 1
512 x 1
Softmax

연구 설계

구현 내용

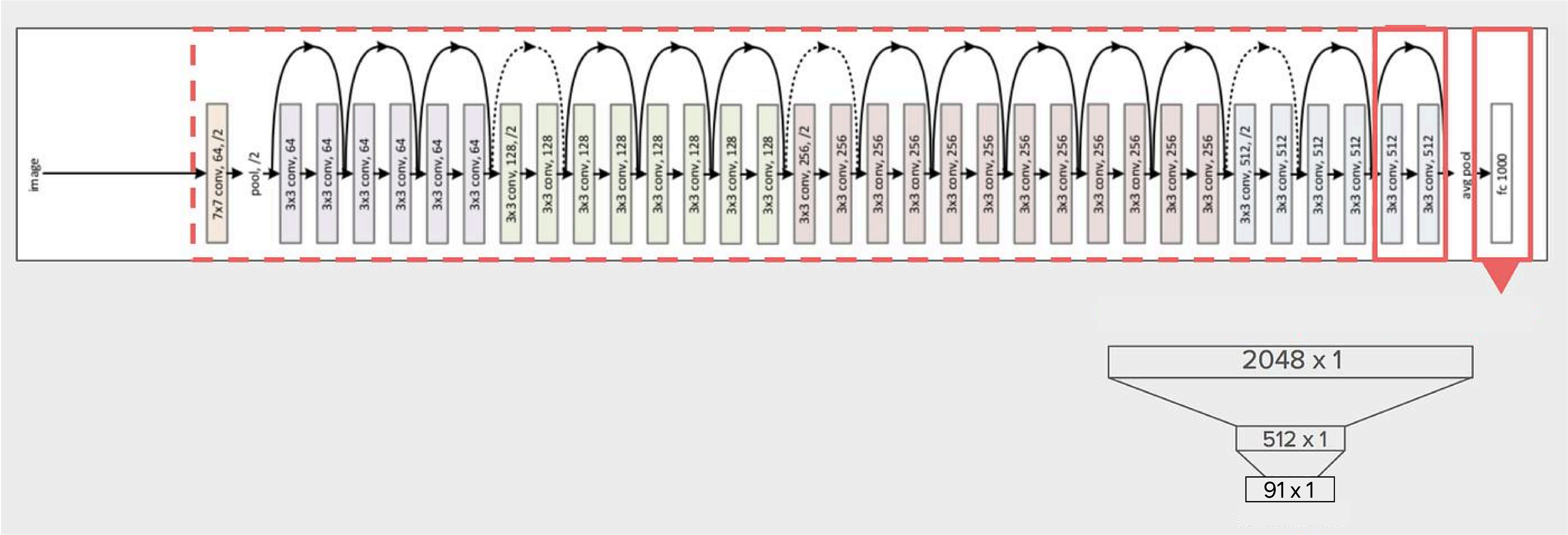


image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
avg pool
fc 1000
2048 x 1
512 x 1
91 x 1

구현 내용

Grad

Grad-CAM++

Grad-Loss = 

\+

\+

CE Loss

$$\mathrm{CE} = -\sum_{c=1}^{C} y_{i,c} \cdot \log(p_{i,c})$$

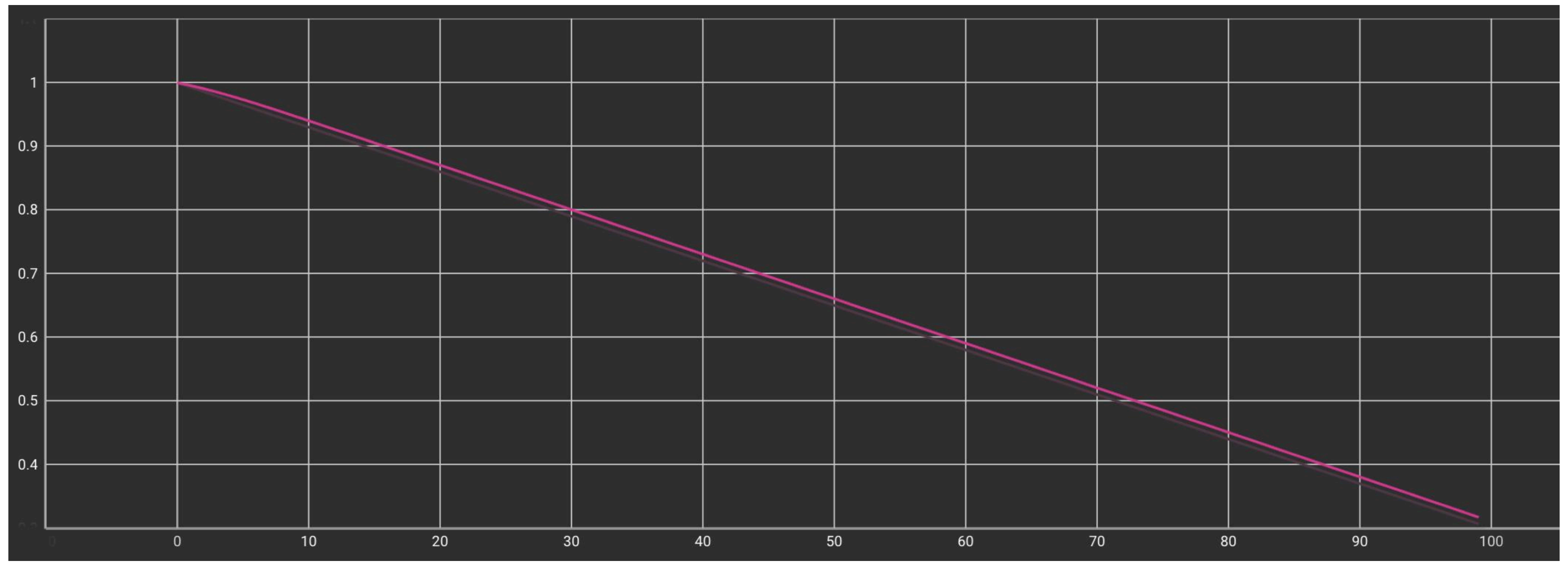
구현 내용

$$L_{total} = \underset{\substack{\uparrow \\ \text{CE Loss 반영률}}}{\lambda_{ce}} \cdot \overset{\substack{\text{CE Loss} \\ \downarrow}}{L_{ce}} + \underset{\substack{\uparrow \\ \text{Grad-CAM Loss 반영률}}}{\lambda_{grad}} \cdot \overset{\substack{\text{Grad-CAM Loss} \\ \downarrow}}{L_{grad}}$$

구현 내용

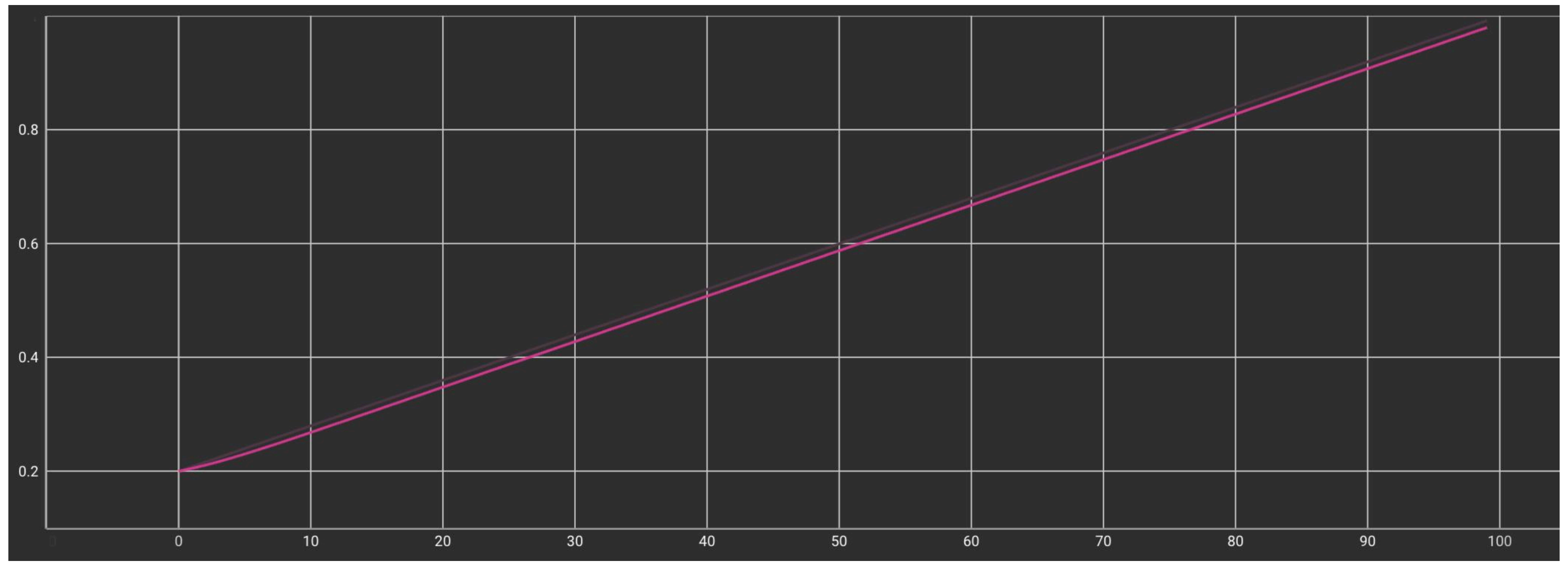
$$\lambda_{ce} = initial\lambda_{ce} + (final\lambda_{ce} - initial\lambda_{ce}) \cdot \frac{current\_epoch}{total\_epoch}$$

Lambda_ce



$$\lambda_{grad} = initial\lambda_{grad} + (final\lambda_{grad} - initial\lambda_{grad}) \cdot \frac{current\_epoch}{total\_epoch}$$

Lambda_grad



구현 내용

설정값:

$$initial\lambda_{ce} = 1.0,\ final\lambda_{ce} = 0.3,$$

$$initial\lambda_{grad} = 0.2,\ final\lambda_{grad} = 1.0$$

구현 내용

L_{grad} =

Grad-CAM++

\+

구현 내용

IoU 계산

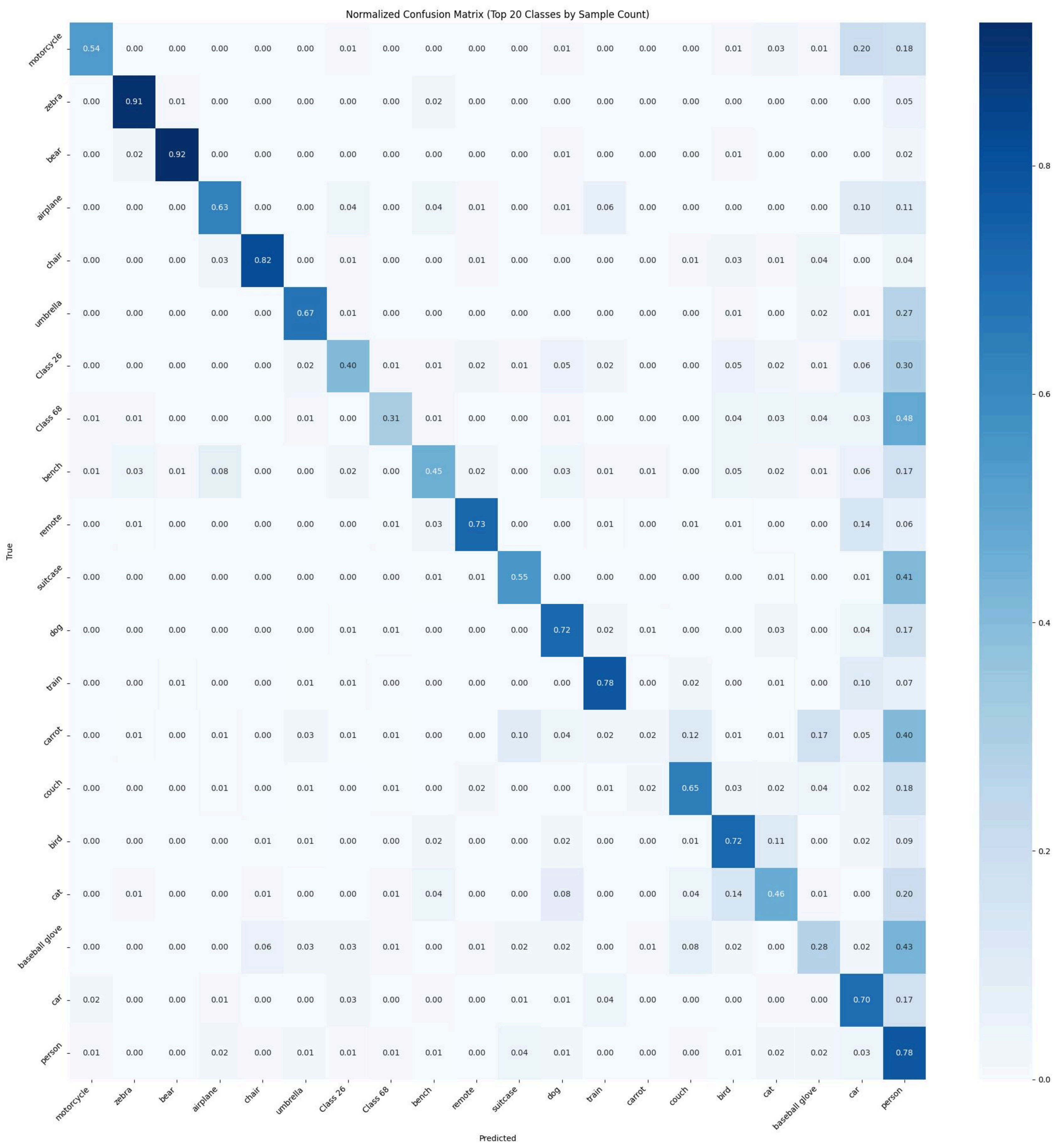
L_{grad} =

\+

λ_{obj} = 1.0

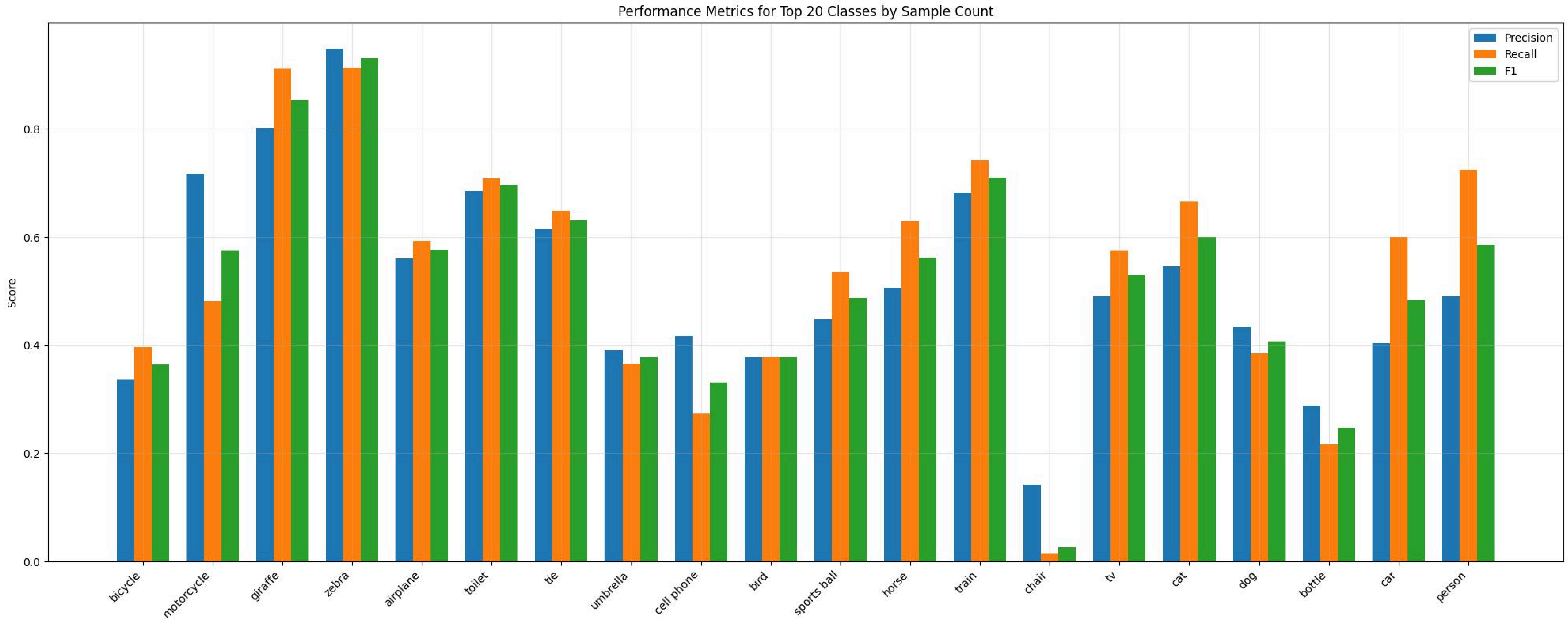
λ_{noobj} = 0.5

결론


Normalized Confusion Matrix (Top 20 Classes by Sample Count)
True
Predicted
motorcycle
zebra
bear
airplane
chair
umbrella
Class 26
Class 68
bench
remote
suitcase
dog
train
carrot
couch
bird
cat
baseball glove
car
person
0.0
0.2
0.4
0.6
0.8

Grad-Loss

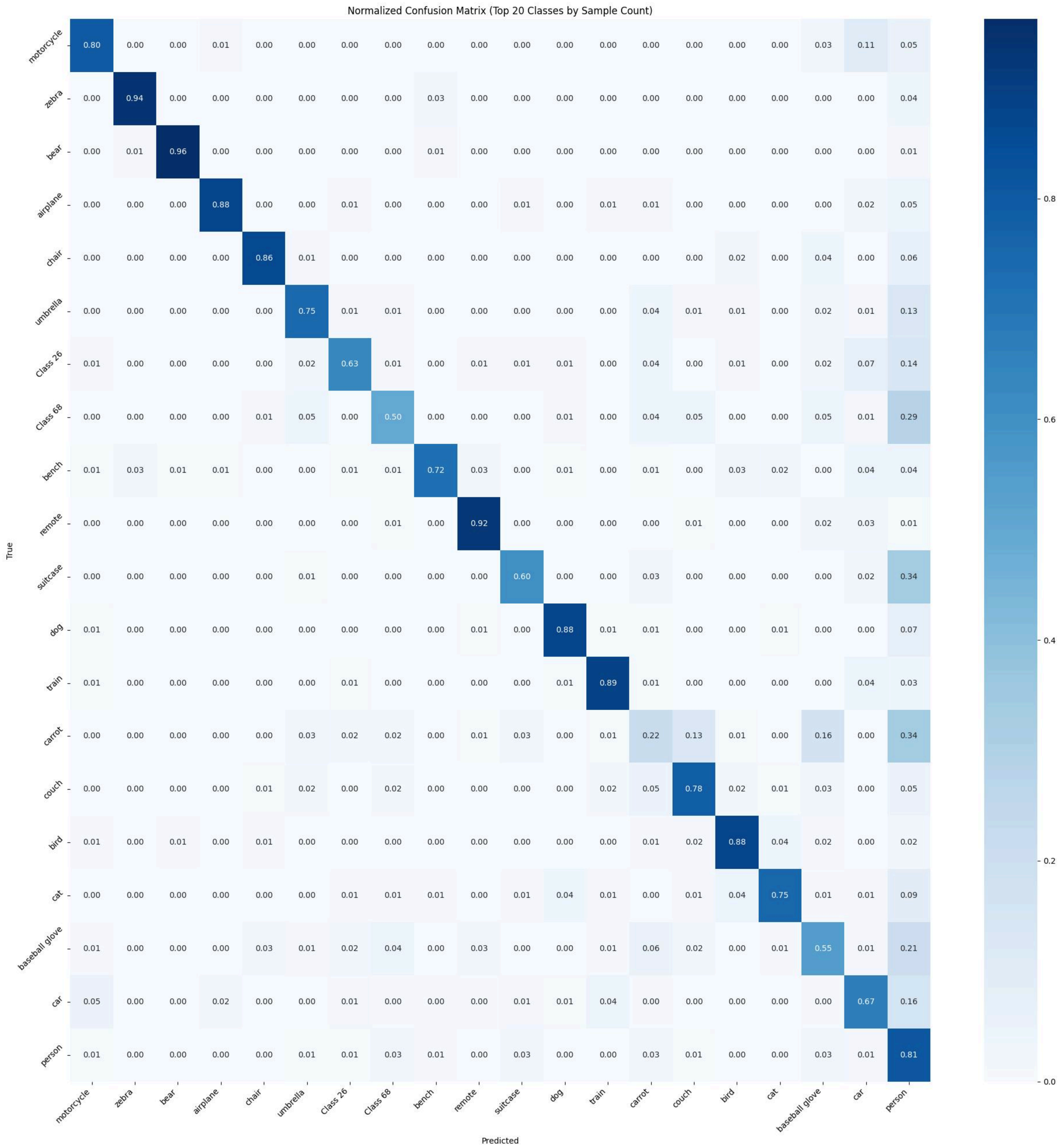
결론



Performance Metrics for Top 20 Classes by Sample Count
Precision
Recall
F1
Score
0.0
0.2
0.4
0.6
0.8
bicycle
motorcycle
giraffe
zebra
airplane
toilet
tie
umbrella
cell phone
bird
sports ball
horse
train
chair
tv
cat
dog
bottle
car
person

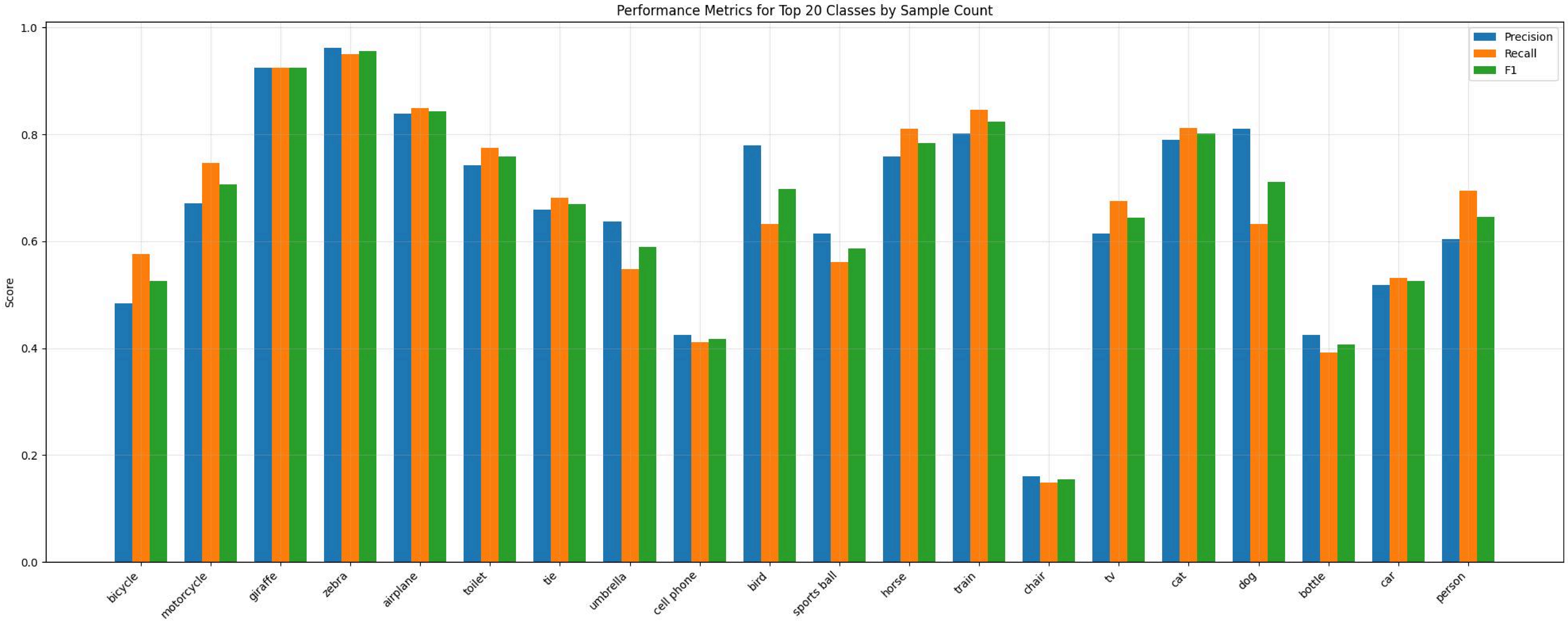
Grad-Loss

결론


Normalized Confusion Matrix (Top 20 Classes by Sample Count)
True
Predicted
motorcycle
zebra
bear
airplane
chair
umbrella
Class 26
Class 68
bench
remote
suitcase
dog
train
carrot
couch
bird
cat
baseball glove
car
person

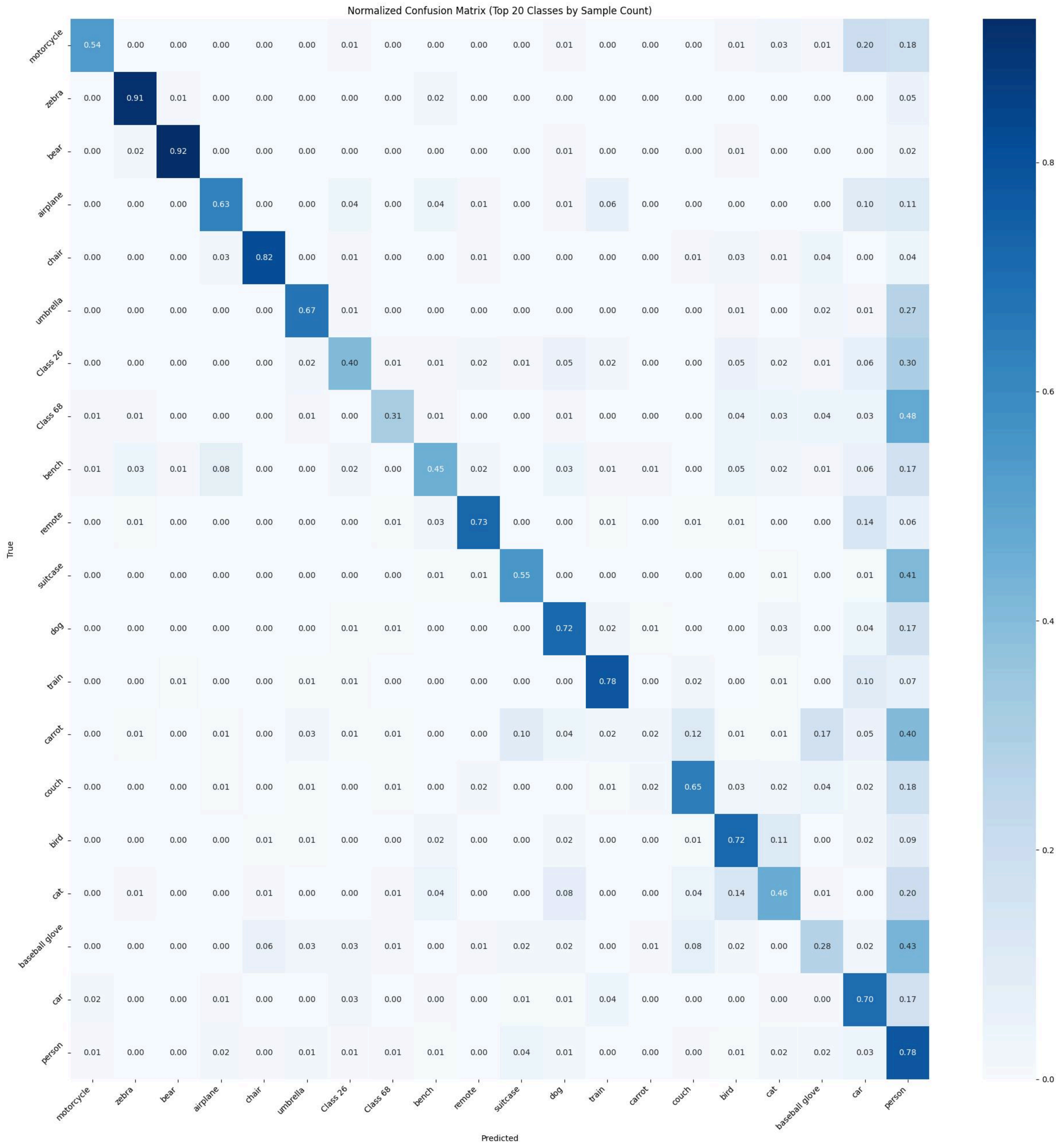
CE-Loss

결론

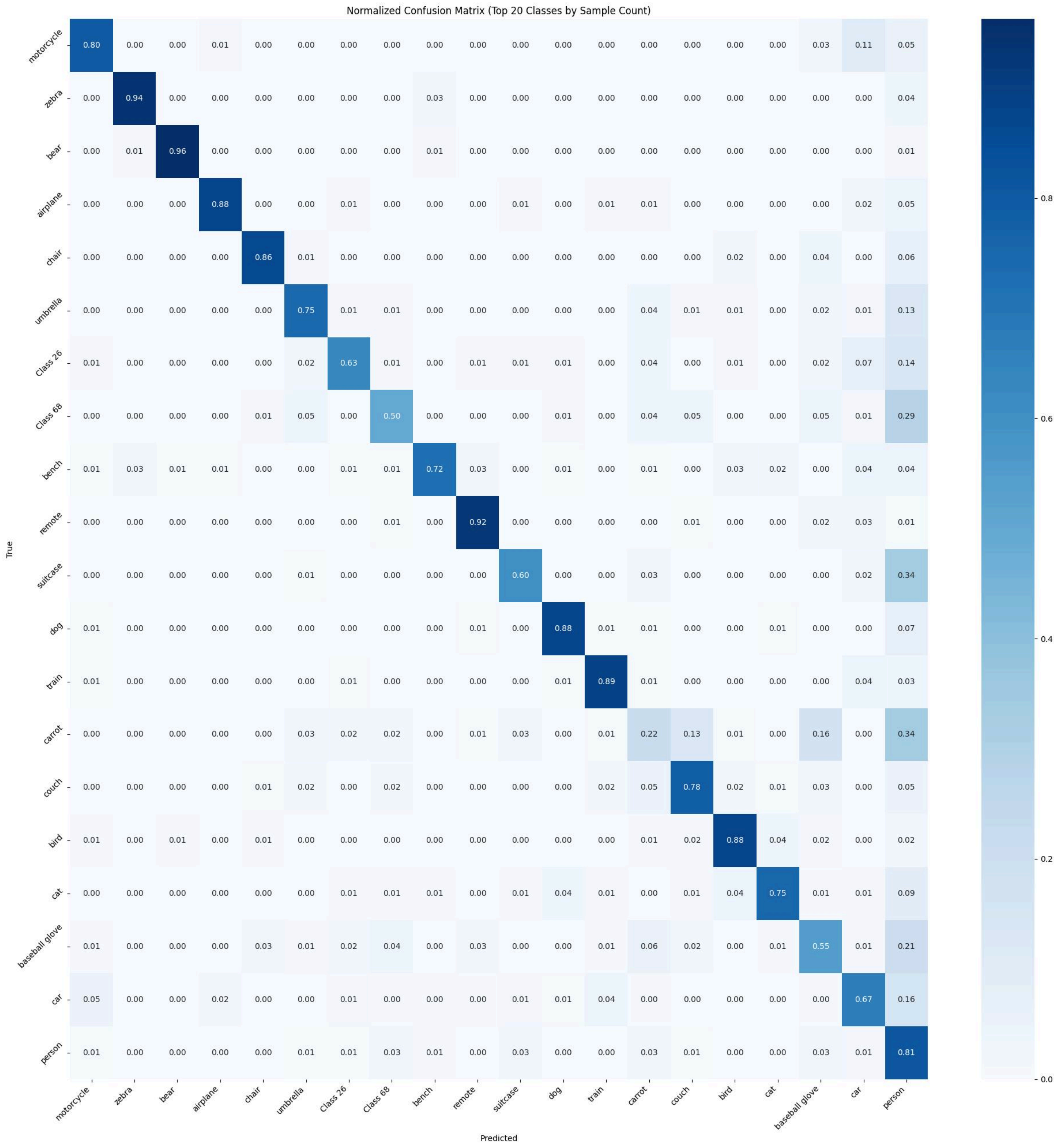

Performance Metrics for Top 20 Classes by Sample Count
Score
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
bicycle
motorcycle
giraffe
zebra
airplane
toilet
tie
umbrella
cell phone
bird
sports ball
horse
train
chair
tv
cat
dog
bottle
car
person

CE-Loss

결론

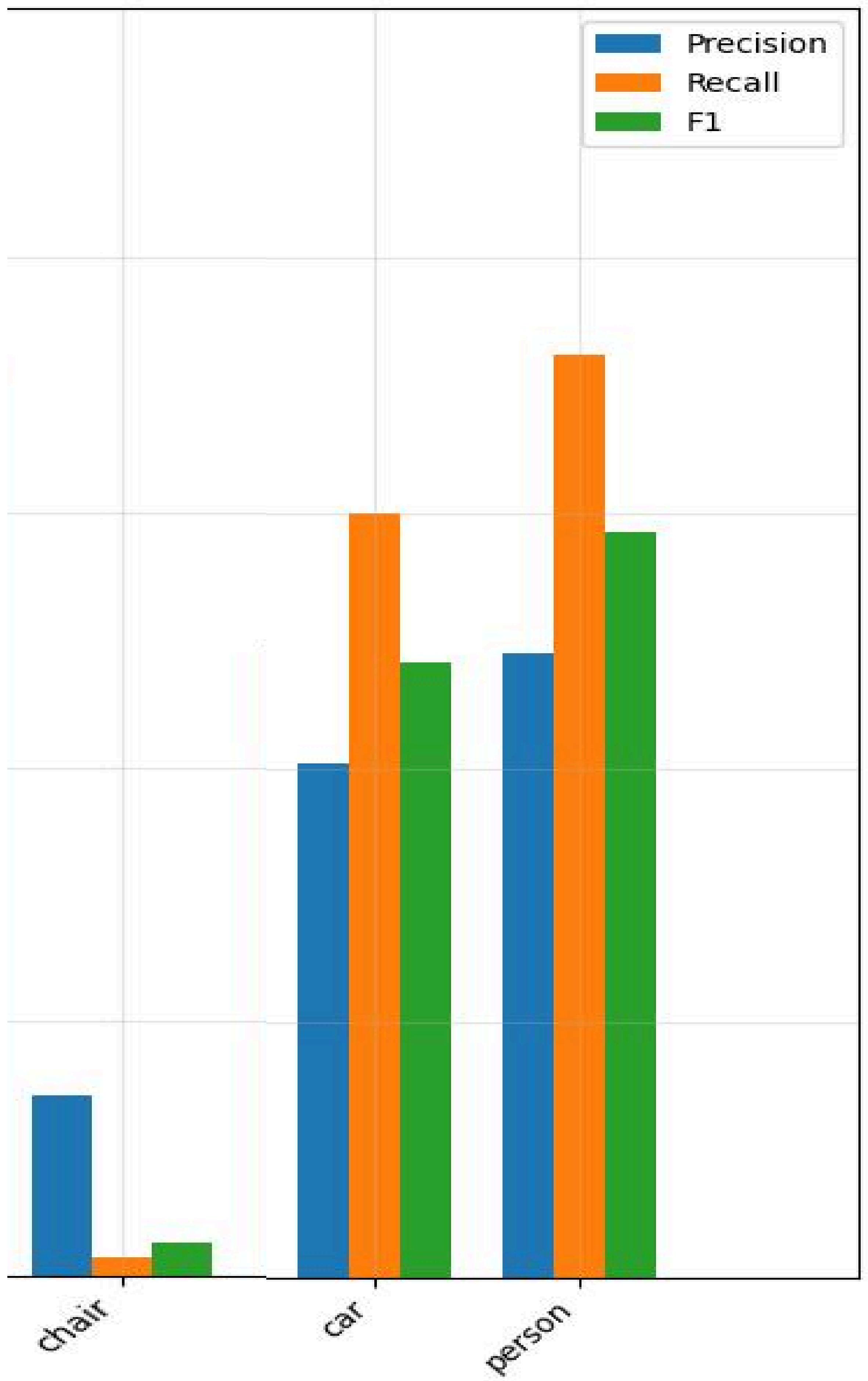


Grad-Loss

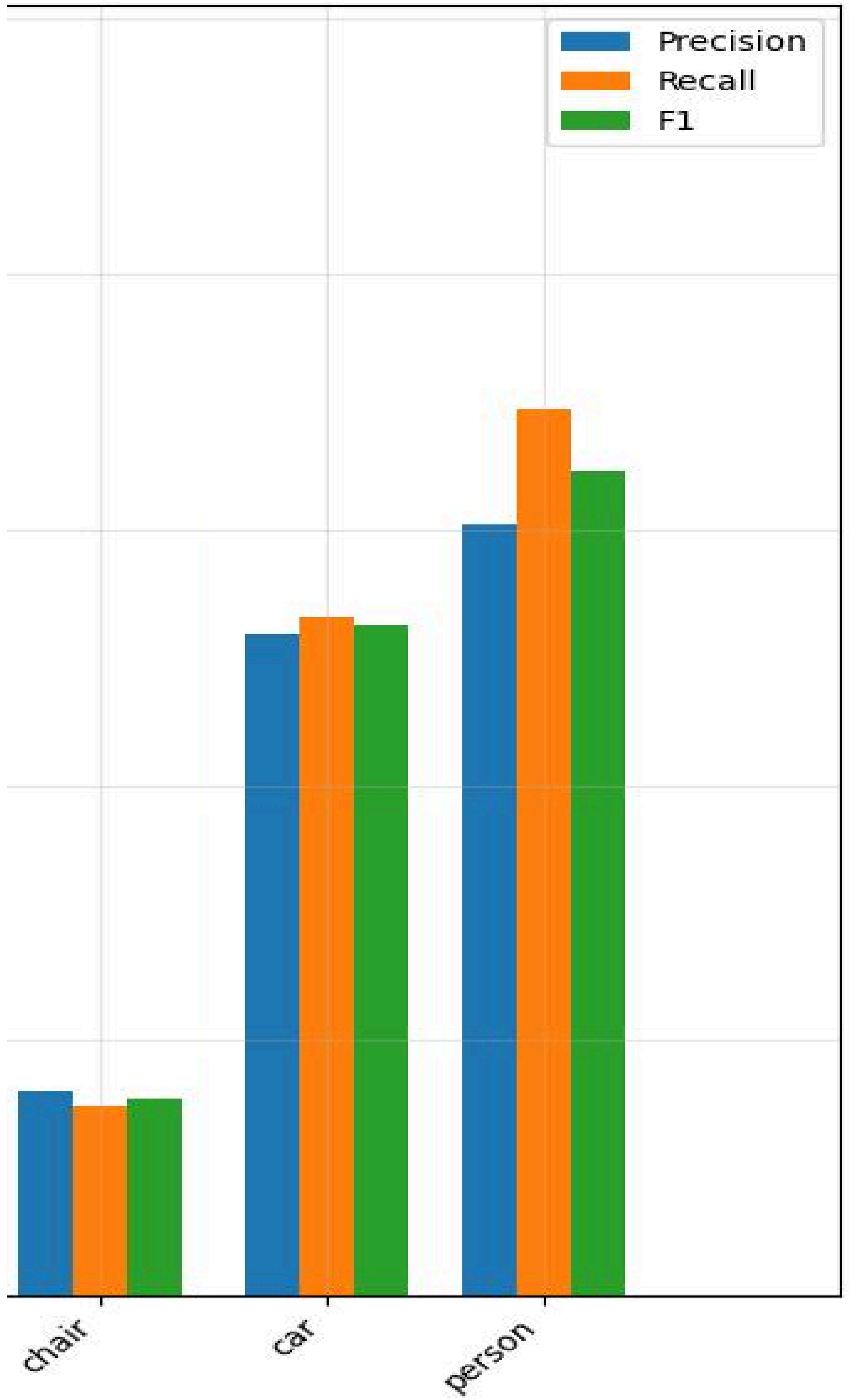


CE-Loss

결론



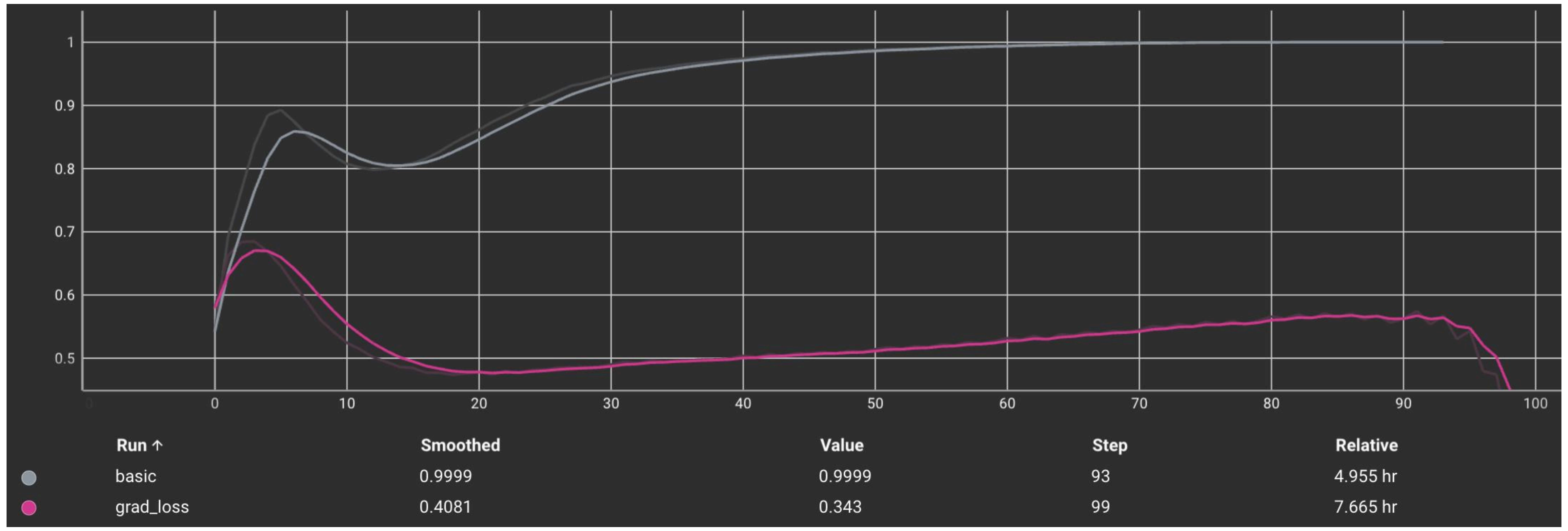
Grad-Loss



CE-Loss

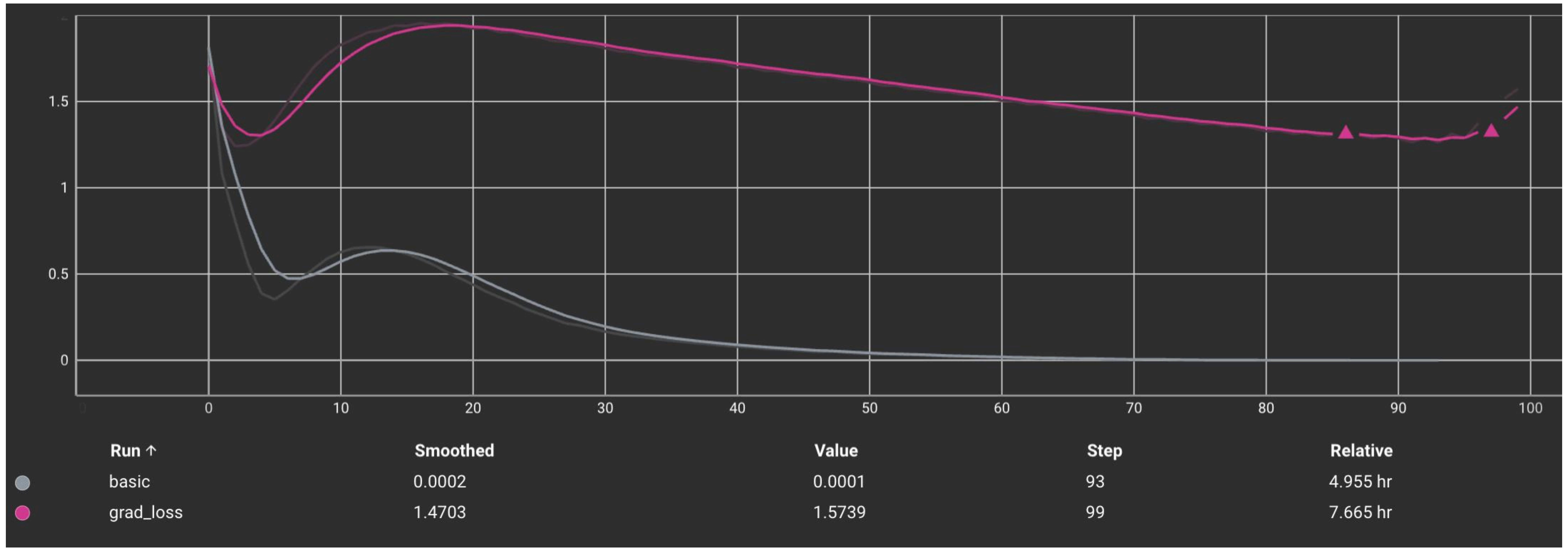
결론

Acc



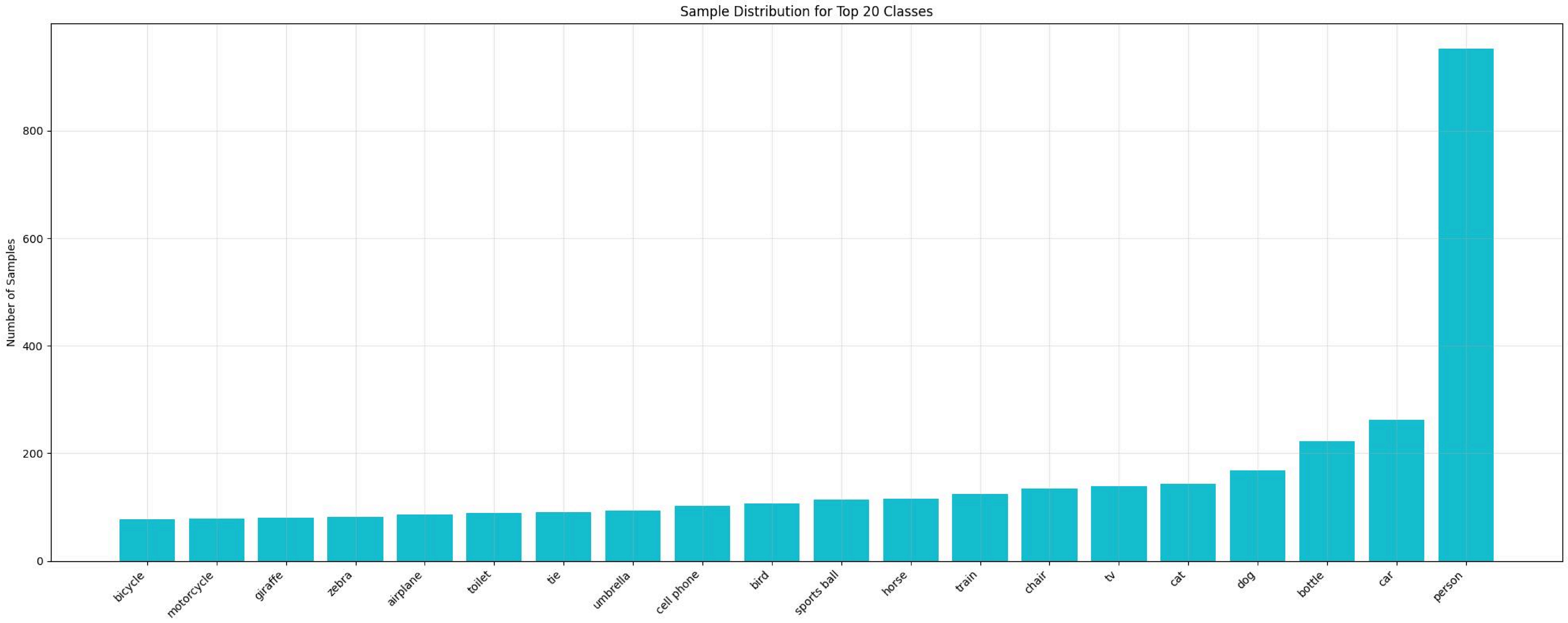
1
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
60
70
80
90
100
Run ↑ Smoothed Value Step Relative
basic 0.9999 0.9999 93 4.955 hr
grad_loss 0.4081 0.343 99 7.665 hr

Loss



1.5
1
0.5
0
0
10
20
30
40
50
60
70
80
90
100
Run ↑ Smoothed Value Step Relative
basic 0.0002 0.0001 93 4.955 hr
grad_loss 1.4703 1.5739 99 7.665 hr

결론


Sample Distribution for Top 20 Classes
Number of Samples
0
200
400
600
800
bicycle
motorcycle
giraffe
zebra
airplane
toilet
tie
umbrella
cell phone
bird
sports ball
horse
train
chair
tv
cat
dog
bottle
car
person

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